

Problem Statement

The objective of this project is to evaluate the practical performance and efficiency of the **YOLOv12n** object detector in comparison to its predecessors, **YOLO11n** and **YOLOv8n**. The goal is to quantify the accuracy-speed tradeoffs introduced by newer architecture designs in real-time scenarios.

- **Expected Input:** Digital images from the COCO 2017 validation split (5,000 images).
- **Desired Output:** Precise classification and localization of objects across 80 categories, provided as bounding boxes with associated confidence scores and class labels.

Approach

The project utilizes an **attention-centric design** philosophy introduced in YOLOv12, which aims to enhance feature representation without significantly increasing computational overhead. The technical implementation leverages the **Ultralytics v8.3.63** framework. All models are evaluated at a standardized 640px input size on a Tesla T4 GPU to ensure a fair comparison of inference latency and detection accuracy (AP).

Experiments and Results

The experimental setup involved benchmarking the three models on the COCO 2017 validation set using a batch size of 16. Existing Ultralytics evaluation code was exploited for metric calculation, while custom scripts were used to aggregate throughput data.

Key Findings:

- **Accuracy:** YOLOv12n achieved the highest overall accuracy with an **AP50-95 of 0.404**, representing a 1.0 point gain over YOLO11n and a 3.0 point gain over YOLOv8n.
- **Scale Sensitivity:** Improvements were consistent across all object sizes, with the most notable gains observed in large object detection (AP L).
- **Latency:** A success for the project was defined as identifying the accuracy-speed frontier; experiments revealed YOLOv12n has an inference time of **5.1 ms**, compared to ~3.0 ms for older variants, highlighting a deliberate tradeoff for higher precision.

Success is defined by the model's ability to provide superior localization (AP75) and recall (AR 100) while maintaining a throughput suitable for real-time applications on standard GPU hardware.